

40.520 Stochastic Models

Spring 2026 Final Exam

Guideline

- **Topics:** Poisson Processes, Continuous Time Markov Chains, Martingale Theory, ~~Brownian Motion~~
- **Date & Time:** Tuesday, April 28th, 2:00 – 4:00 PM (Papers handed out at 1:55 pm)
- **Location:** Think Tank 22 (Building 2, Level 3, Room 2.311)
- **Permitted Materials:**
 - A Non-programmable scientific calculator
 - One double-sided A4 cheat sheet
- Scratch paper will be available upon request.
- Make-up final exam will be scheduled before May 1st.

Review Points

Poisson Processes

- Exponential Distribution:
 - key properties: sum and minimum of exponential variables, memoryless property
 - use these properties to solve related problems
- Poisson Processes
 - definition and construction
 - stationary and independent increments
- Properties of Poisson Process
 - decomposition and superposition of Poisson processes
 - apply these to solve problems
- Conditional distribution of arrival times
- Time-inhomogeneous Poisson process
 - time-varying intensity function
 - distribution of event counts over a given time interval

- Compound Poisson process:
 - formulate a problem as a compound Poisson process and solve it
 - mean and variance
- ~~Mixed Poisson Processes~~

Continuous Time Markov Chain

- definitions and construction of a CTMC
- transition probabilities and the forward/backward Kolmogorov equations
- transient probabilities
- balance equations and limiting probabilities
- model simple queueing systems using CTMC: M/M/1, M/M/n, with abandonment, with finite waiting room (e.g., M/M/1/k, ...)

Martingale Theory

- Definition of a martingale; determine whether a given stochastic process is or is not a martingale
- Apply the martingale convergence theorem to find the distribution of X_T , where T is a stopping time
- ~~Martingale inequalities and the strong law of large numbers (SLLN)~~

Important Proofs

- Prove that if the interarrival times of a counting process are i.i.d. exponential with rate λ , then the number of events by time t follows a Poisson distribution with mean λt (i.e., the counting process is a Poisson process).
- Prove that the decomposition (splitting) of a Poisson process into two independent Poisson processes, and the superposition (merging) of independent Poisson processes, again yield Poisson processes.
- Derive the mean and variance of a compound Poisson process of the form $X(t) = \sum_{i=1}^{N(t)} Y_i$, where $N(t)$ is Poisson and Y_i are i.i.d. independent of $N(t)$.
- Use the Markov property of a continuous-time Markov chain (CTMC) to prove that the holding time in any state is memoryless (i.e., exponentially distributed).
- Derive the transient probabilities $P_{ij}(t) = P(X(t) = j \mid X(0) = i)$ for a continuous-time Markov chain with two states.